A Micromachined Doubly-Clamped Beam Rheometer for the Measurement of Viscosity and Concentration of Silicon-Dioxide-in-Water Suspensions

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***Abstract*— In this contribution we demonstrate the feasibility of a sensor system for viscosity and concentration measurement of complex liquids, in particular suspensions of silicon dioxide particles in water. The sensor system is based on a doubly clamped micromachined beam vibrating in the sample liquid, and an optical readout utilizing a DVD player pickup head. The vibrating beam features resonance frequencies in the range of several 10 kHz, and higher mechanical amplitudes than microacoustic sensors, e.g., quartz thickness shear mode (TSM) resonators or surface acoustic wave (SAW) devices. We show that the damping of the beam is dominated by the viscosity of the liquid, and that this relation also holds for the considered complex liquids, whereas a TSM resonator sensor fails to detect the steady state shear viscosity of the suspensions.**

I. Introduction

The online measurement of viscosity offers a multitude of possibilities for process or condition monitoring. A common approach to viscosity measurement is the use of microacoustic sensors like TSM resonators or SAW devices. These sensors are characterized by small mechanical amplitudes at high operating frequencies. Liquids showing a more complex structure, e.g., containing particles (suspensions) or droplets (emulsions), need to be deformed significantly compared to their structural size, and must be excited at low frequencies to reveal the rheological parameter of interest [1]. Consequently, the measurement results obtained from microacoustic devices are not directly comparable to those of conventional laboratory equipment.

The design of miniaturized sensors for such applications is therefore crucial. We have recently presented a novel sensor system based on a doubly clamped vibrating beam interacting with the liquid, and an optical readout (Fig. 1), and have demonstrated the feasibility of this setup for the measurement of the viscosity of liquids [2]. The beam's frequency characteristics feature first mode resonance frequencies in the range of several ten kHz, depending on the liquid loading.

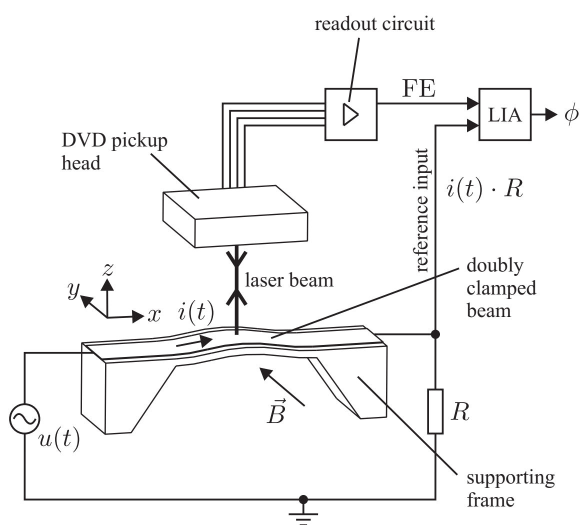

Fig. 1. Schematic diagram of the sensor system [2]. The beam vibrations are excited by Lorentz forces caused by the excitation current $i(t)$ and the magnetic flux density $\vec{B}$. The phase shift ϕ between the excitation current and the beam deflection in vertical (z-) direction is obtained by an optical readout based on a DVD player pickup head, and is subsequently used to determine the resonance frequency and the damping of the beam.

Measurements using a variety of alcohols have shown that the damping factor D (the reciprocal of the quality factor Q) of such a beam is dominantly influenced by the viscosity of the respective liquid. In this contribution, we aim at demonstrating the applicability of the sensor system for the measurement of viscosity and particle concentration of suspensions of SiO_2 particles in H_2O, showing complex rheological behavior.

II. Sample Liquids

As a model system for liquids exhibiting complex rheological behavior, silicon–dioxide–in–water suspensions have

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TABLE I

SiO_2 IN H_2O SUSPENSIONS WITH DIFFERENT PARTICLE VOLUME CONCENTRATIONS Φ

	Concentration Φ	Starting solution [ml]	DI water [ml]	dynamic viscosity η [mPa·s]	mass density ρ [kg/m^3]
Sample 1	0.224	50	0	7.74	1290
Sample 2	0.2108	47	3	6.65	1273
Sample 3	0.2019	45	5	5.64	1261
Sample 4	0.1794	40	10	4.08	1232
Sample 5	0.1480	33	17	2.94	1191
Sample 6	0.1211	27	23	2.21	1156
Sample 7	0.493	11	39	1.23	1062
Sample 8	0	0	50	0.932	998

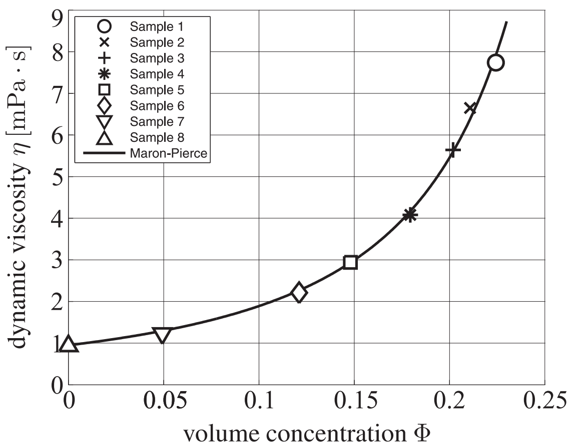

Fig. 2. Dynamic viscosity η of the sample liquids (suspensions of SiO_2 in H_2O), measured using a cone-plate rheometer. The relationship between viscosity and particle volume concentration Φ can be described by the Maron-Pierce model (solid line), cf. (1).

been used. A solution with a particle volume concentration of 22.4 % (silicon dioxide amorphous gel, 40 % weight concentration, ammonium stabilizing counter ion, average particle size 20–22 nm, from ABCR GmbH & Co. KG, Karlsruhe, Germany) has been diluted with deionized (DI) water. The liquid quantities and concentrations of the mixtures are given in Tab. I. Figure 2 depicts the dynamic viscosity η of the suspensions at ambient temperature, determined using a Brookfield LV-DV+CP cone–plate rheometer. The results reveal, that the relationship between the particle volume concentration Φ and η is properly modeled by the Maron–Pierce model [3], [4],

$$\eta = \eta_0 \left(1 - \frac{\Phi}{\Phi_{\max}}\right)^{-2}, \tag{1}$$

with $\eta_0 = 0.9475\,\mathrm{mPa \cdot s}$, and $\Phi_{\max} = 0.3431$. In Fig. 2, this model is represented by the solid line. The mass densities of the suspension samples have been calculated from the respective mixing ratios, and the density values of the starting solution and of DI-water, $1290\,\mathrm{kg/m^3}$ and $998\,\mathrm{kg/m^3}$, respectively (Tab. I).

For comparison, a second set of liquids has been used. The pure chemicals ethanol, 1-propanol, 1-butanol, 1-pentanol, 1-hexanol, 1-heptanol, and 1-octanol cover approximately the same viscosity range as the considered suspensions, but show Newtonian behavior. They exhibit a constant dynamic viscosity, independent of the shear rates applied to them. The dynamic viscosities and mass densities of these liquids have been extracted from the literature [5], [6].

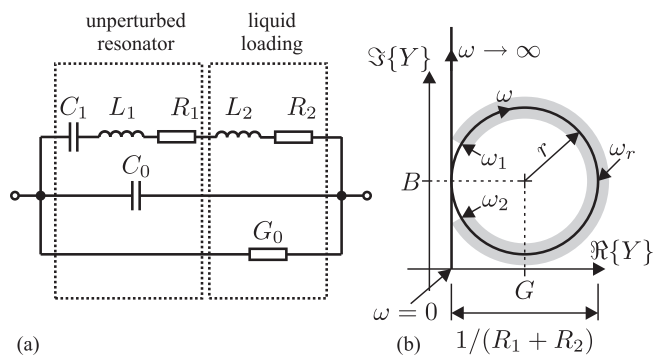

Fig. 3. TSM quartz resonator measurements. (a) Lumped element equivalent circuit of the quartz resonator immersed in a conductive liquid. (b) Sensor admittance locus plot $Y(\mathrm{j}\omega)$: In the vicinity of the resonance frequency, the quartz admittance can be approximated by a circular arc with the radius r, and the center $G + \mathrm{j}B$.

III. Quartz Resonator Measurements

For the quartz TSM resonator measurements, an AT-cut quartz resonator has been used, exhibiting a fundamental shear mode resonance frequency of 6 MHz. The sensor features a quartz diameter of 8 mm and an electrode diameter of 4 mm. When the resonator is immersed in a liquid, its behavior in the vicinity of the first mode resonance frequency can be modeled by the lumped element equivalent circuit of Fig. 3a. The motional arm R_1–L_1–C_1 represents the mechanical resonance and the internal losses, and C_0 accounts for the static capacitance of the device. Since both electrodes are in contact with the sample liquid, and the suspensions examined in this work are electrically conductive, the liquid's conductivity must be considered by G_0. The acoustic load, i.e., the liquid surrounding the TSM quartz resonator, is modeled by R_2 and L_2. To first order, both R_2 and L_2 are proportional to the square root of the viscosity–density product $\sqrt{\eta\rho}$ [7].

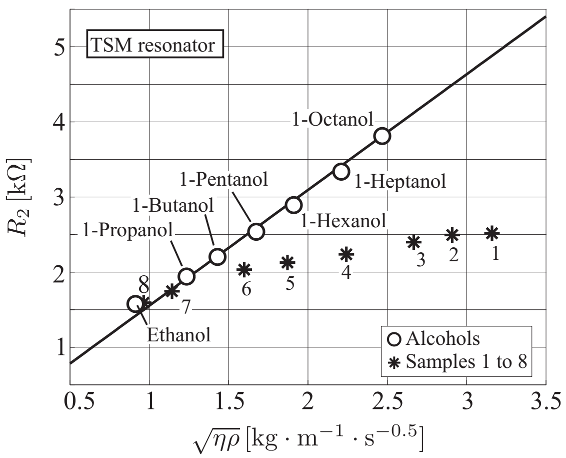

Fig. 4. Equivalent circuit resistance R_2 (Fig. 3a) of the quartz TSM resonator (6 MHz, 8 mm diameter AT-cut), immersed in the sample liquids, vs. square root of the viscosity–density product of the respective liquid. The results indicate that a TSM resonator is not suitable for the measurement of viscosity or concentration of the considered SiO_2–in–H_2O suspensions (Sample 1 to 8), whereas the results for Newtonian liquids (alcohols) follow $R_2 \sim \sqrt{\eta\rho}$.

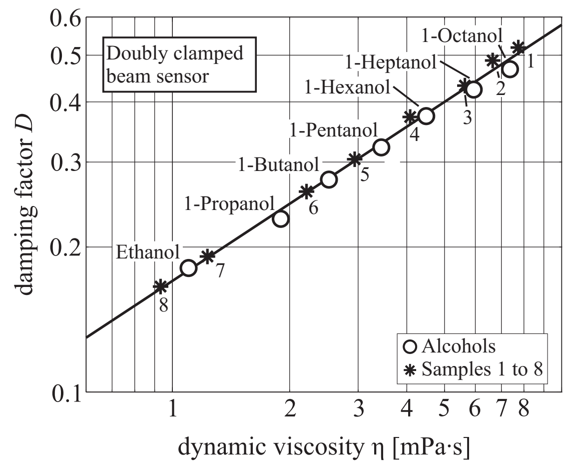

Fig. 5. Measurement results demonstrating the feasibility of the presented viscosity sensor for use in complex liquids. The figure shows good agreement between the damping factor D obtained from the sensor system and the results of conventional viscosity measurement, η. Small deviations from the curve fit can be attributed to the influence of the liquid's differing mass densities.

The sensor was immersed in the respective liquid, and the resonator admittance Y was measured using a HP 8753E network analyzer, yielding $Y(\mathrm{j}\omega_k)$, where ω_k are the angular frequencies covered by the frequency sweep measurement. In principle, the expression for the quartz admittance deduced from Fig. 3a,

$$Y(\mathrm{j}\omega) = \frac{1}{R_1 + R_2 + \frac{1}{\mathrm{j}\omega C_1} + \mathrm{j}\omega(L_1 + L_2)} + \mathrm{j}\omega C_0 + G_0, \quad (2)$$

could be fitted to the $Y(\mathrm{j}\omega_k)$ to determine the values of R_2 and L_2, but due to the number of circuit elements involved this fit only converged for thoroughly chosen starting values. In the vicinity of the resonance frequency, $\omega_1 < \omega_k < \omega_2$, the admittance locus plot of the quartz can be approximated by a circular arc (Fig. 3b). The radius r of this arc is given by $r = 1/[2(R_1 + R_2)]$, and its center is denoted by $G + \mathrm{j}B$. Therefore, the value of $R_1 + R_2$ can be obtained by minimizing the objective function

$$J(G, B, r) = \sum_k \left(|Y(\mathrm{j}\omega_k) - G - \mathrm{j}B| - r\right)^2 \quad (3)$$

with respect to G, B, and r [8]. For the liquids considered in this work, R_1 is negligible compared to R_2.

Figure 4 compares the measurement results obtained for both sets of sample liquids described in section II. As expected, the equivalent circuit resistance R_2 of the quartz resonator immersed in the alcohols, which show Newtonian behavior, is proportional to the square root of the viscosity–density product, $\sqrt{\eta\rho}$. However, the results obtained for the suspensions, do not follow this trend. The values of R_2 indicate, that the effective viscosities probed by the quartz resonator are lower than those determined by conventional laboratory viscometers. This behavior corresponds to the results previously obtained for emulsions [1].

IV. Doubly Clamped Beam Sensor

A schematic diagram of the doubly clamped beam sensor system is given in Fig. 1. The sensor device has been formed by coating a silicon wafer with a stack of silicon nitride layers, and a titanium–gold–chromium layer. Subsequently, the doubly clamped thin beam has been formed by back-side KOH etching and reactive ion etching from the front side [9]. The beam geometries are $320\times40\times1.3\,\mu\mathrm{m}^3$. The beam vibrations are excited by placing the device in the magnetic flux density $\vec{B}$ (in y-direction) of a permanent magnet, and driving a sinusoidal current through the beam's conducting layer. To obtain the resonance frequency ω_r and the damping factor D of the beam, the beam deflection must be determined. In the sensor system, an optical readout based on a laser pickup head of a DVD player is utilized [10]. A detailed description of the device fabrication, the measurement setup, and the optical readout method can be found in [2], [9].

The sensor device was immersed in the sample liquid (sample volume approximately 1 ml), and an excitation current of 3.1 mA rms was driven through the beam's conducting path. The magnetic flux density provided by the permanent magnet is approximately 100 mT. A frequency sweep measurement yields the phase shift $\phi(\mathrm{j}\omega)$ between the excitation current $i(t)$ and the actual beam deflection. To these measurement results, the transfer function of a second order system,

$$\phi(\mathrm{j}\omega) = \arg\left\{\frac{1}{1 + \mathrm{j}\omega\frac{2D}{\omega_\mathrm{r}} - \frac{\omega^2}{\omega_\mathrm{r}^2}}\right\}, \quad (4)$$

was fitted, resulting in the resonance frequency ω_r and the damping factor D.

The damping factor D obtained for the vibrating beam immersed in the sample liquids is plotted in Fig. 5. As already presented in [2], the damping factor D is dominantly influenced by the liquid's viscosity η. In contrast to the measurement results obtained by the TSM quartz resonator (Fig. 4), the doubly clamped beam device exhibits similar behavior for both sets of liquids. The measured damping factors as a function of the viscosity appear on the same trend line for both the Newtonian liquids and the suspensions (solid line in Fig. 5). Small deviations can be attributed to the influence of the liquid's differing mass densities.

The evaluation of the measurement results also yields the resonance frequency $f_\mathrm{r} = \omega_\mathrm{r}/(2\pi)$. For the alcohol sample liquid set it has been shown, that f_r is influenced by both the liquid's dynamic viscosity and mass density [2]. In principle, using a suitable model for the vibrating beam, the mass density of the liquids could thus be determined simultaneously [11]. However, for the suspensions, the results were widely spread over a range of several kHz, and no clear trend could be observed. This behavior could possibly be attributed to an additional mass loading of the beam due to a contamination with silicon oxide particles, and would require further investigation and special treatment of the measurement results for f_r.

V. Conclusions

Liquids exhibiting complex rheological behavior must be deformed significantly and with sufficiently low frequencies to reveal a viscosity parameter comparable to that obtained from conventional laboratory viscometers. In this work, we have used suspensions of silicon dioxide in water as a model system for such complex liquids. A thickness shear mode quartz resonator operating at a frequency of several MHz and low vibration amplitudes probes these fluids in a different rheological regime compared to a laboratory cone-plate rheometer. The results of this microacoustic sensor do not follow the expected viscosity–density product dependence. In contrast, the damping factor obtained from a doubly clamped beam sensor with a resonance frequency in the range of several ten kHz and vibration amplitudes in the nanometer range is comparable to the dynamic viscosity probed by the laboratory viscometer.

Since the relationship between the viscosity and the particle volume concentration of the considered silicon–dioxide–in–water suspensions is properly described by the Maron-Pierce model, the doubly clamped beam sensor can also be used to determine the concentration of the observed suspensions.

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